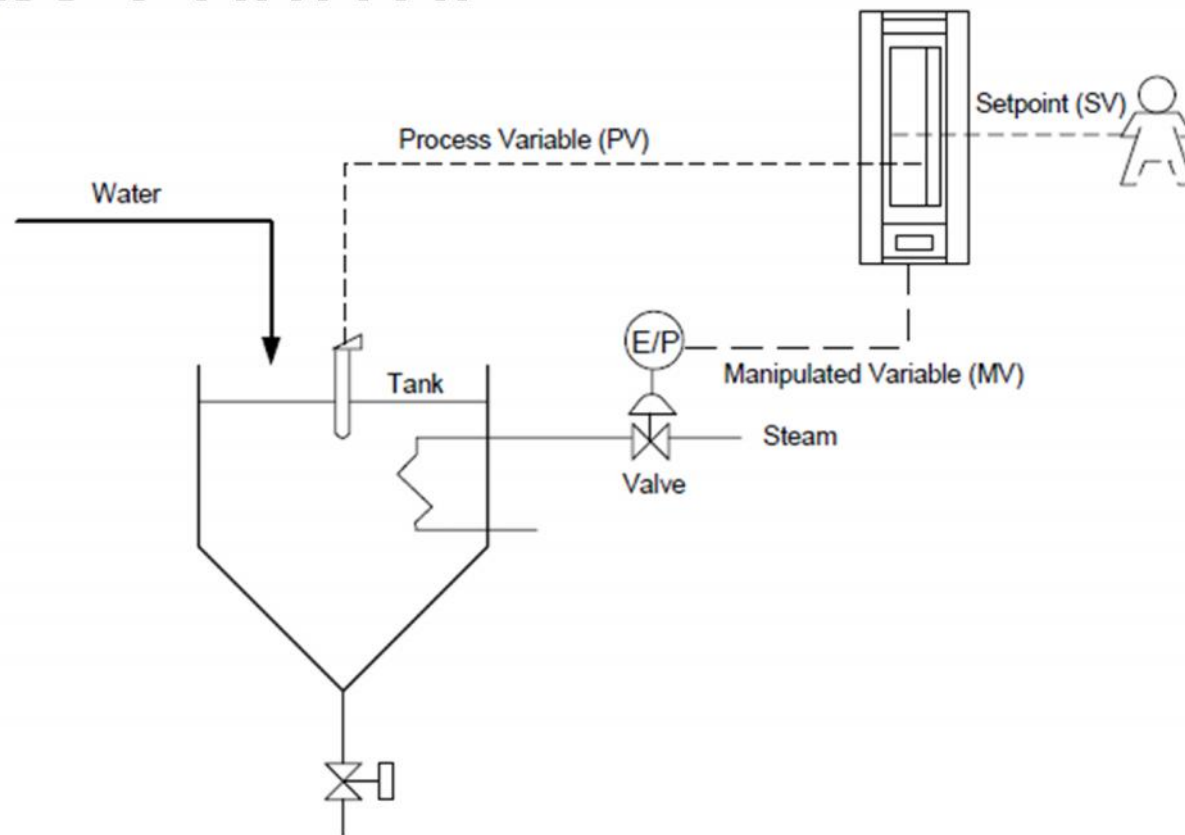


DCS Yokogawa Centum CS3000

M. Yazid

9, 16, 23 October 2012

Process Control



DCS

- DCS merupakan sistem kontrol yang mampu menghimpun (mengakuisisi) data dari lapangan dan memutuskan akan diapakan data tersebut, secara singkat DCS -> ambil/baca data + lakukan pengontrolan berdasar data tersebut. Data-data yang telah diakuisisi (diperoleh) dari lapangan bisa disimpan untuk rekaman atau keperluan-keperluan masa datang, atau digunakan dalam proses-proses saat itu juga, atau bisa juga, digabung dengan data-data dari bagian lain proses, untuk kontrol lanjutan dari proses yang bersangkutan

Jenis-jenis kontrol

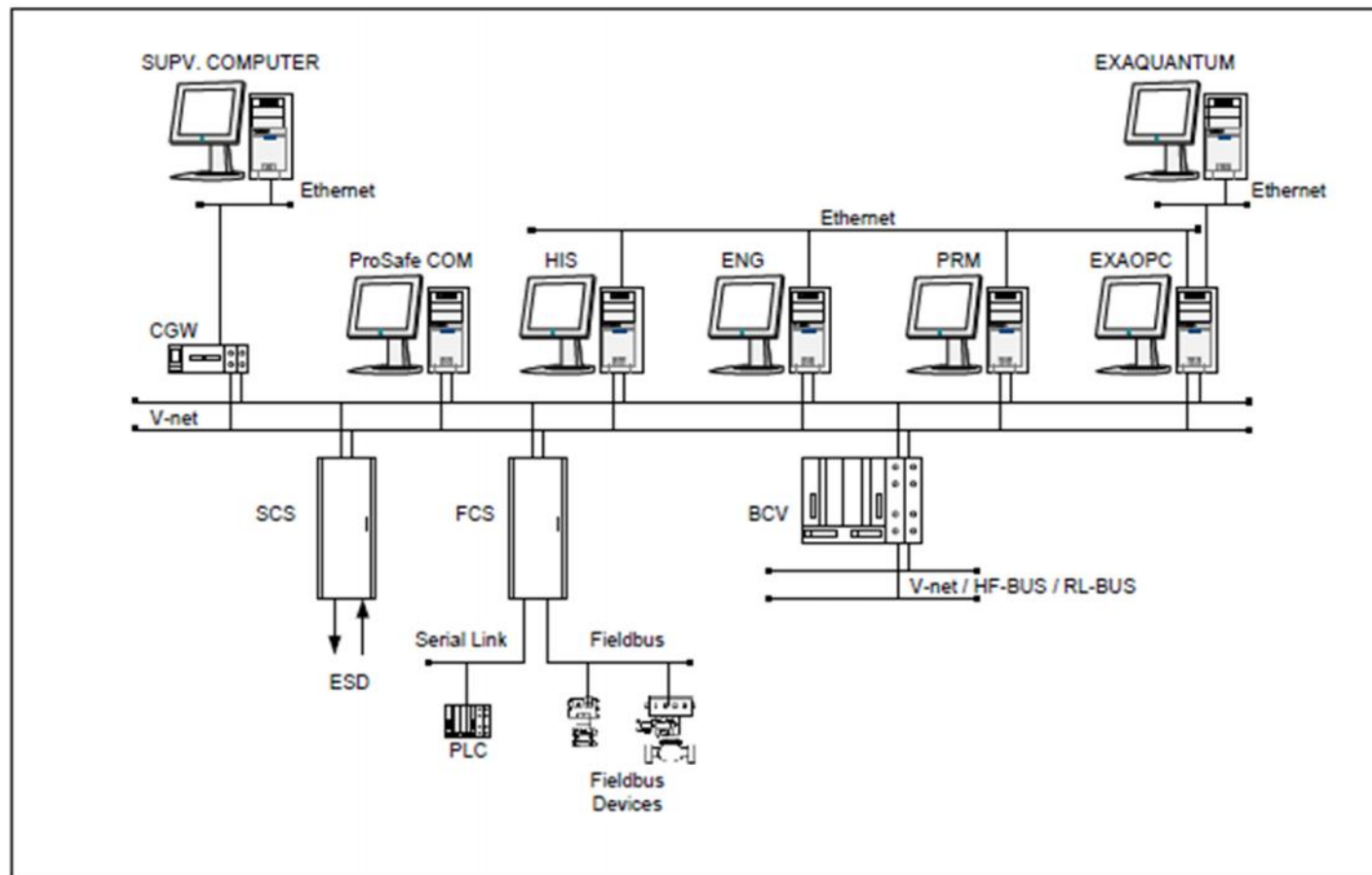
- ♦ **Feedback Control**
Control that acts to correct the process variable (e.g. Temperature in a tank) to agree with the target value (setpoint) by comparing both.
- ♦ **Feed forward Control**
Control which takes a corrective action by measuring the disturbances (e.g. Ambient temperature) and directly driving the valve before it affects the process.
- ♦ **Sequential Control**
Control that successively advances each control step in accordance with the pre-determined sequence.



Komponenten DCS

- EWS (Engineering Workstation) & HIS (Human Interface Station)
- FCS (Field Control Station)
- V-net : Real time control network

Konfigurasi Sistem DCS



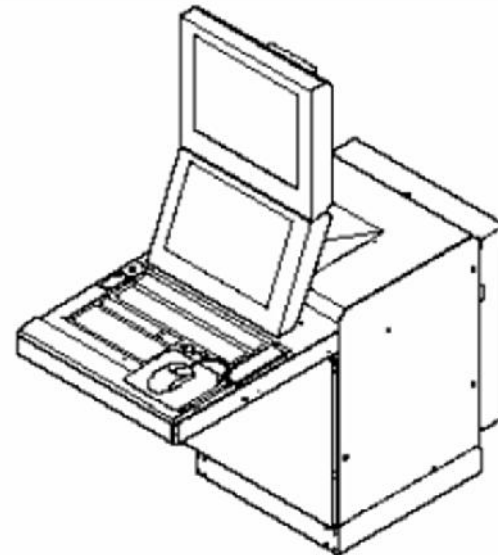
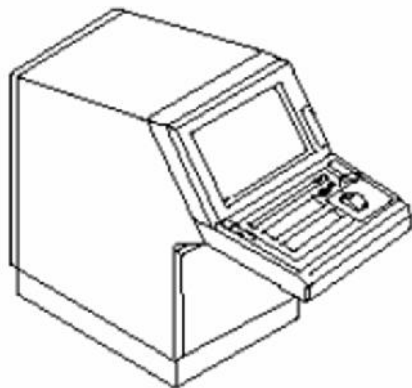


Komponen Hardware

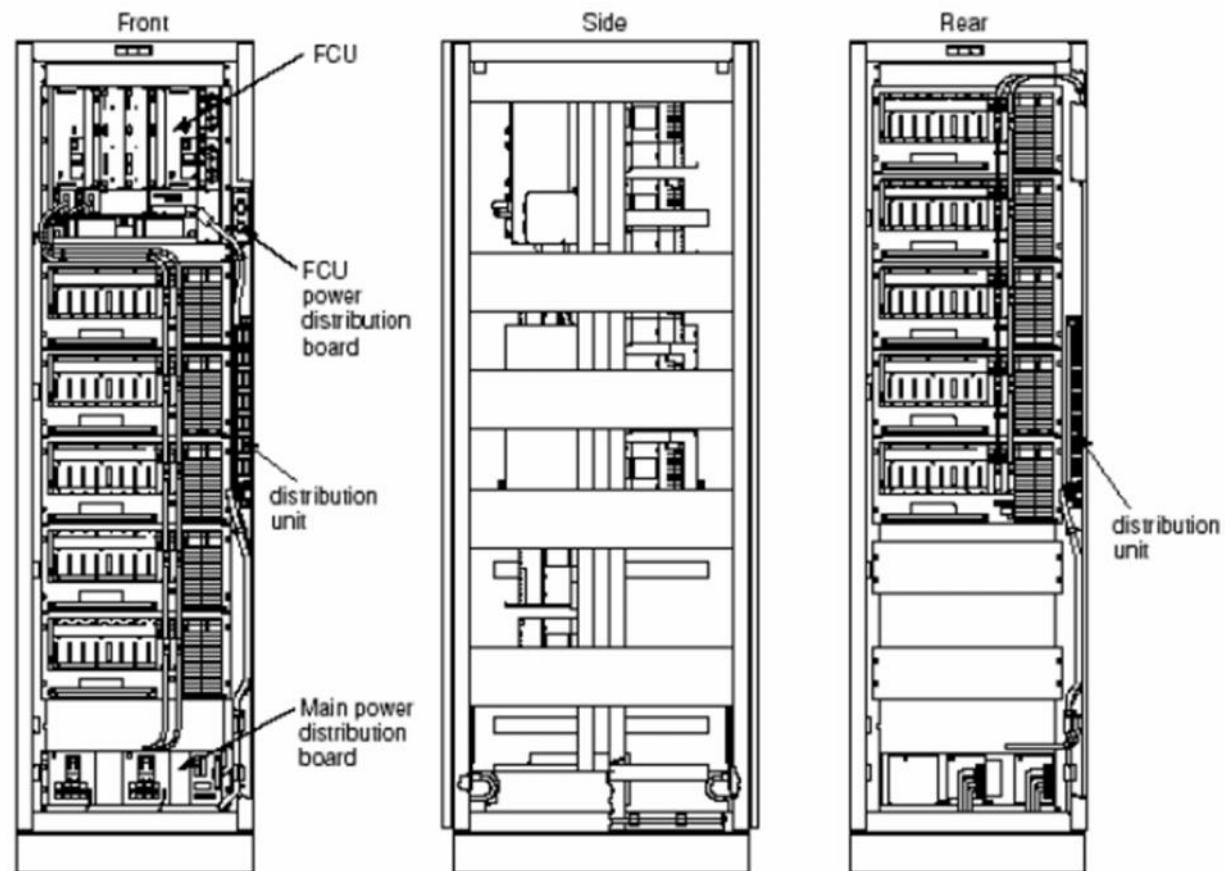
- **HIS (Human Interface System)**
Adalah unit yang digunakan operator untuk operasi dan monitoring proses plant
- **EWS (Engineering Workstation)**
Adalah unit yang dipakai untuk konfigurasi dan modifikasi software DCS
- **FCS (Field Control System)**
Adalah konponen yang melakukan semua kontrol dan perhitungan proses input dan output dari lapangan

HIS

- Tipe tertutup
- Tipe terbuka

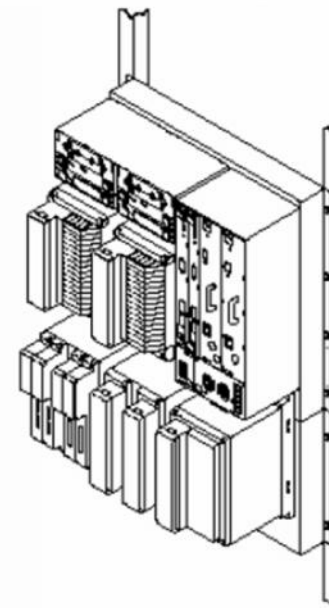
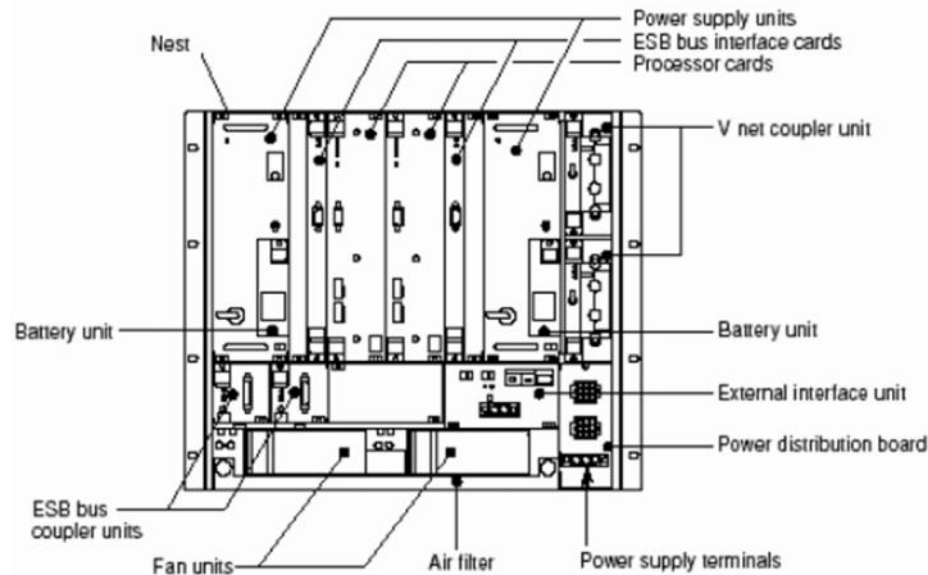
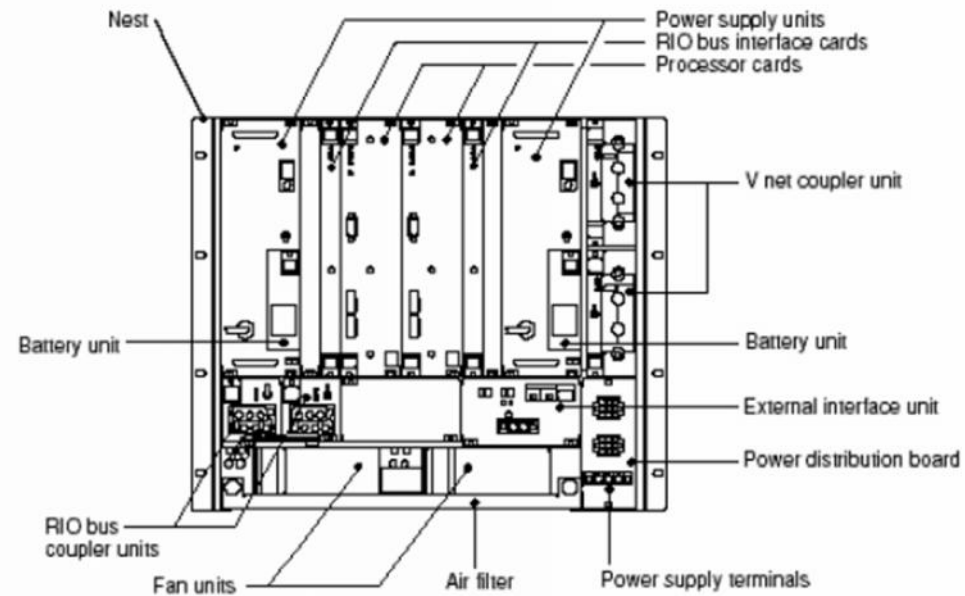


FCS



FCS

- KFCS
- LFCS
- PFCD

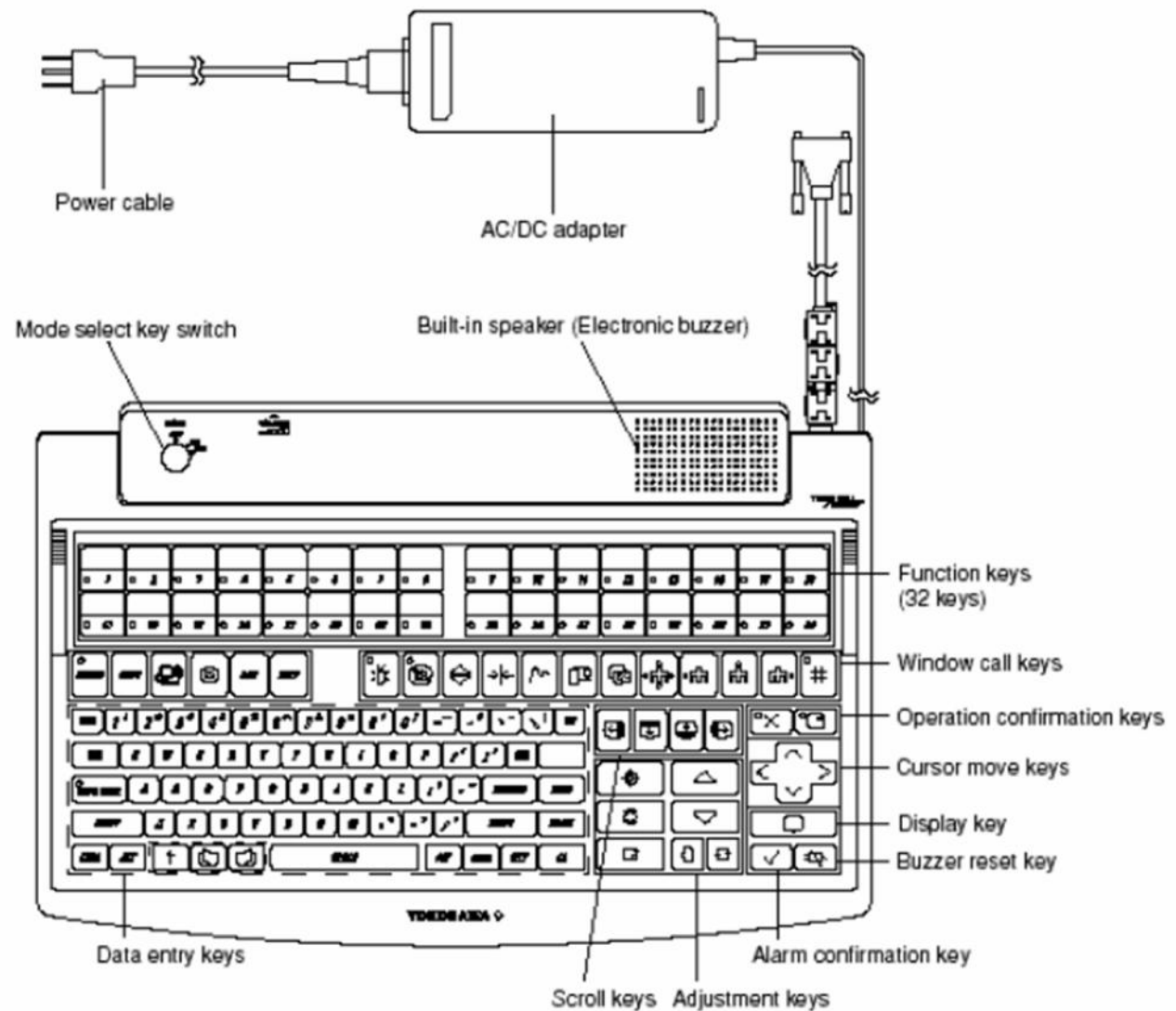




FCS Hardware

- FCU
- Control Bus Unit
- Power Supply Unit
- Back Up Batteries
- RIO
- PIO
- IOU
- NIU
- Node
- ESB Bus Coupler
- Processor card

Keyboard

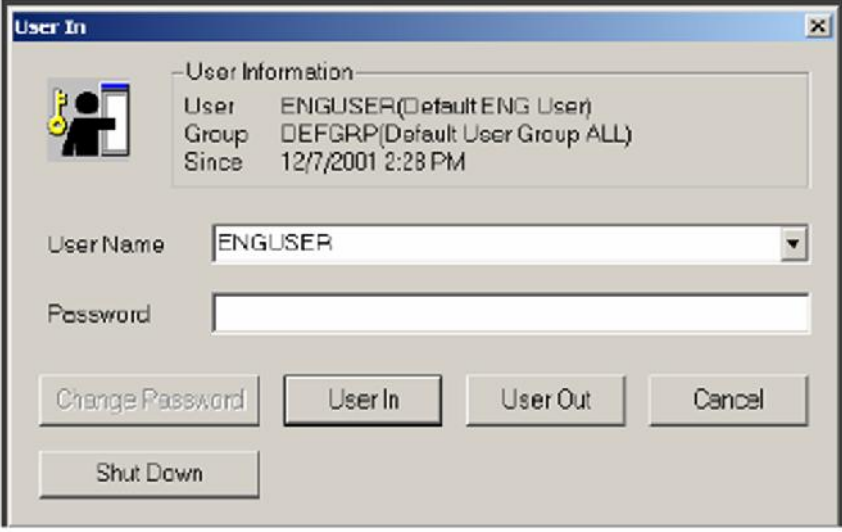


System Message Window



	Process Alarm		Tool Bar
	System Alarm		Navigator Window
	Operator Guide		Window Name
	Message Monitor		Circulate
	User Login		Clear Screen
	Window Call Menu		Buzzer Rest
	Operation Menu		Hard Copy
	Preset Menu		

User Log-in



The image shows a Windows-style dialog box titled "User In". It features a "User Information" section with a small icon of a person with a key. The information displayed is: User: ENGUSER(Default ENG User), Group: DEFGRP(Default User Group ALL), and Since: 12/7/2001 2:28 PM. Below this is a "User Name" dropdown menu showing "ENGUSER" and a "Password" text field. At the bottom, there are five buttons: "Change Password", "User In", "User Out", "Cancel", and "Shut Down".

User Information	
User	ENGUSER(Default ENG User)
Group	DEFGRP(Default User Group ALL)
Since	12/7/2001 2:28 PM

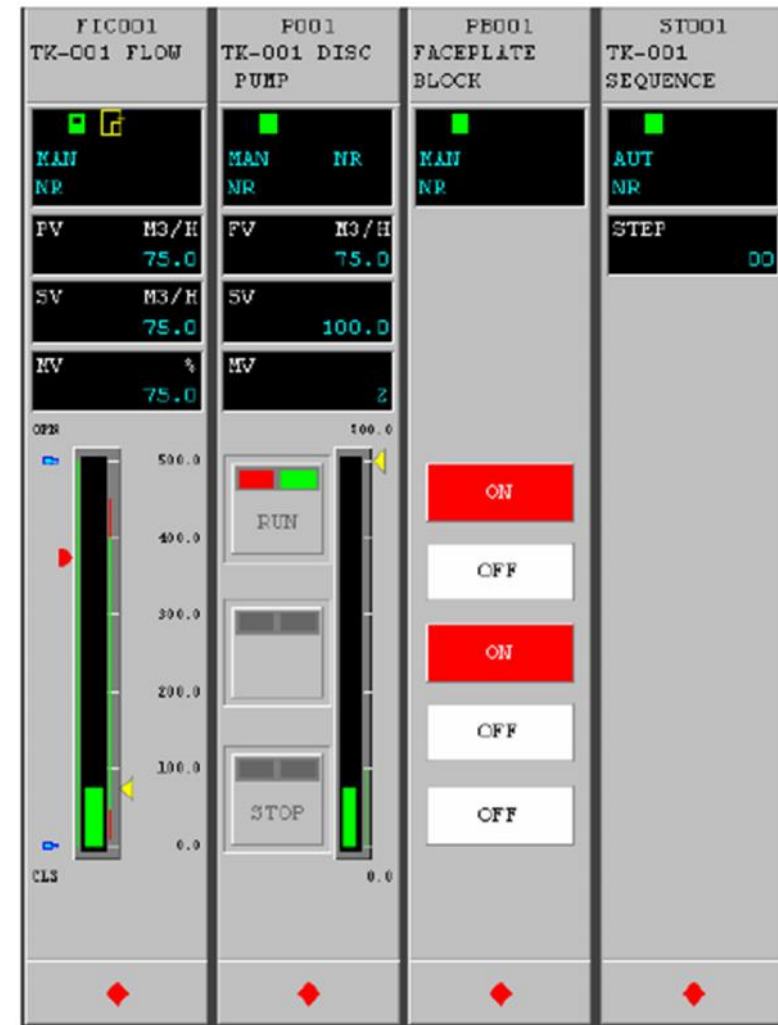
User Name:

Password:

Buttons:

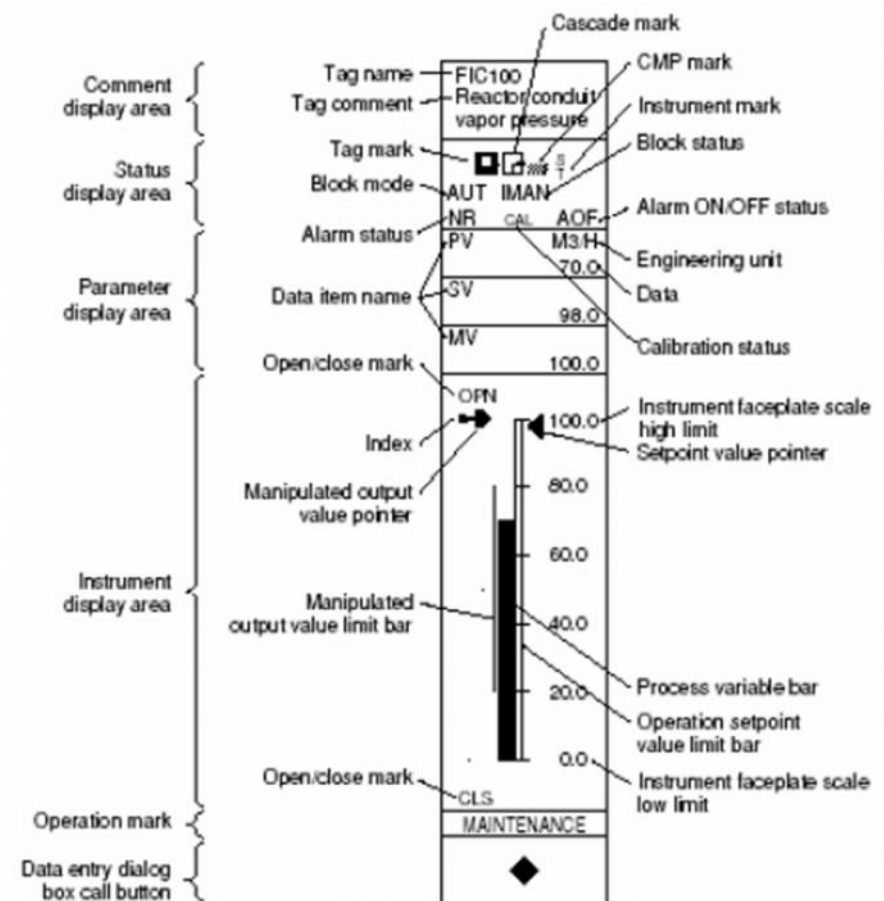
Instrument Faceplate

- Type Analog
- Type Diskret/Digital
- Type Faceplate block
- Type Data Display



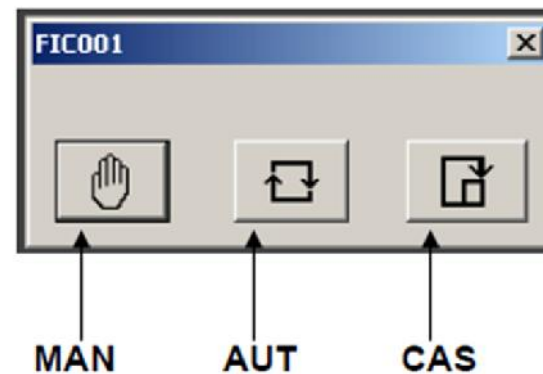
Instrument Faceplate

- Comment Display area
- Status Display Area
- Parameter Display Area
- Instrument Display Area
- Operation Mark
- Data Entry Dialog box call button

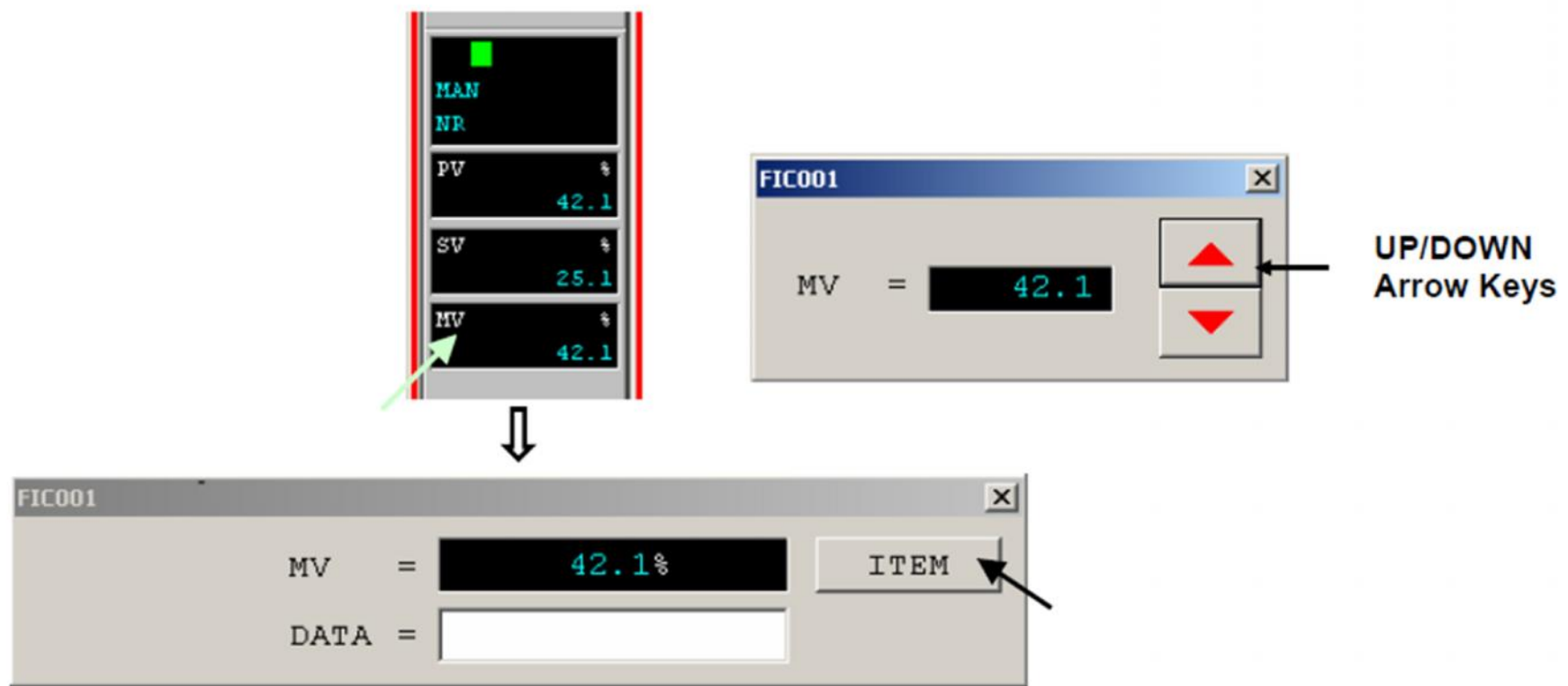


Operation block Mode

- MAN : Manual Mode
- AUT : Auto Mode
- CAS : Cascade Mode



Data Entry Operation



Tuning Window



Figure: To call a Tuning window

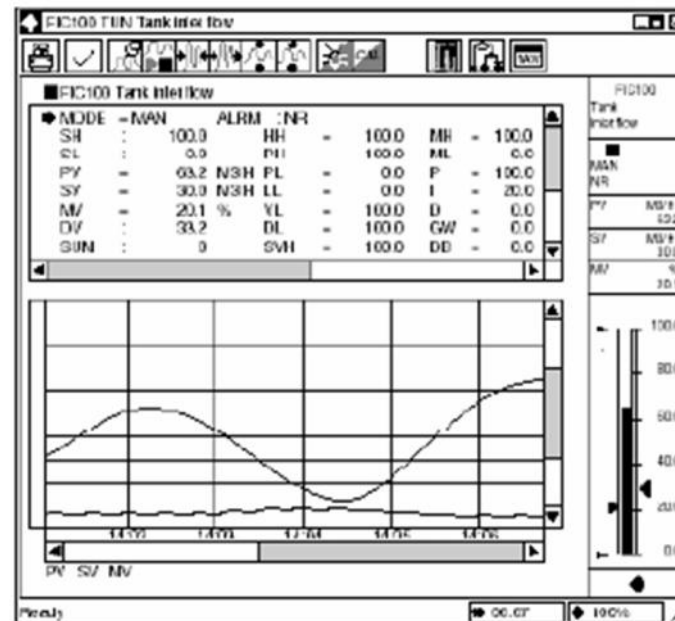


Figure: Tuning window

Tuning Window

Data Item	Data Name	Entry Permitted or Not	Range	Default
MODE	Block mode	X	---	O/S (MAN)
ALRM	Alarm status		---	
PV	Process variable	Δ (*1)	PV engineering unit value	SL
SV	Setpoint value	Δ (*2)	Value in the same engineering unit as PV	SL
MV	Manipulated variable	Δ (*3)	MV engineering unit value	MSL
DV	Control deviation value		Value in the same engineering unit as PV	0
SUM	Totalizer value	X	Engineering unit value	0
OPHI	Output high-limit index	X	MSL to MSH	MSH
OPLO	Output low-limit index	X	MSL to MSH	MSL
MSH	MV scale high limit		Value in the same engineering unit as MV	---
MSL	MV scale low limit		Value in the same engineering unit as MV	---
HH	High - high limit alarm setpoint	X	SL to SH	SH
PH	High - limit alarm setpoint	X	SL to SH	SH
PL	Low - limit alarm setpoint	X	SL to SH	SL
LL	Low - low limit alarm setpoint	X	SL to SH	SL
VL	Velocity alarm setpoint	X	+(SH-SL)	SH-SL
DL	Deviation alarm setpoint	X	+(SH-SL)	SH-SL
SVH	Setpoint high limit	X	SL to SH	SH
SVL	Setpoint low limit	X	SL to SH	SL
MH	Manipulated variable high-limit setpoint	X	MSL to MSH	MSH
ML	Manipulated variable low-limit setpoint	X	MSL to MSH	MSL
P	Proportional band	X	0 to 1000%	100%
I	Integral time	X	0.1 to 10,000 seconds	20 seconds
D	Derivative time	X	0.1 to 10,000 seconds	0 second
GW	Gap width	X	0 to (SH - SL)	0.0
DB	Deadband	X	0 to (SH - SL)	0.0
CK	Compensation gain	X	-10.000 to 10.000	1.000
CB	Compensation bias	X	---	0.000
PMV	Preset manipulated output value	X	MSL to MSH	MSL

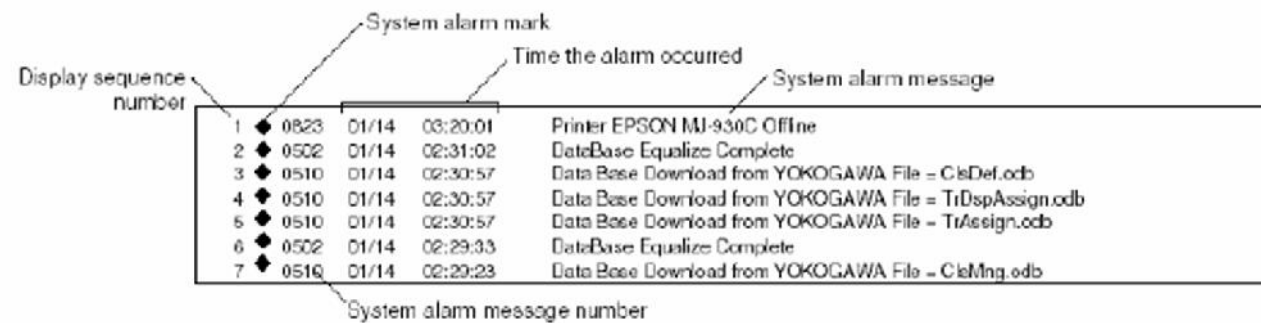
Alarm Status

Symbol	Name	Description
NR	Normal	Indicates a state in which no alarm has occurred.
OOP	Output OPen Alarm	Indicates a state in which the output data status has become output failure (PTPF) as a result of the failure or disconnection of an operation terminal or process I/O device or the abnormality of output destination data. Normally, the output function is stopped.
IOP	High Input Open Alarm	Indicates a state in which the input data status has become bad value (BAD) as a result of the failure or disconnection of a detection terminal or process I/O device or the abnormality of input destination data. Normally, any processing that uses input signals is stopped. If the input signal has been overshoot due to disconnection, etc., this alarm indicates a state in which input is overshoot to the high-limit direction.
IOP-	Low Input Open Alarm	Indicates a state in which the input signal has been overshoot to the low-limit direction due to disconnection, etc. The input data status becomes bad value (BAD). Normally, any processing that uses input signals is stopped.
HH	High High Alarm	Indicates a state in which the process variable exceeds the high high-limit alarm setpoint.
LL	Low Low Alarm	Indicates a state in which the process variable falls below the low low-limit alarm setpoint.
HI	High Alarm	Indicates a state in which the process variable exceeds the high-limit alarm setpoint.
LO	Low Alarm	Indicates a state in which the process variable falls below the low-limit alarm setpoint.
DV+	Deviation Alarm +	Indicates a state in which the deviation between the process variable and the setpoint value exceeds the deviation alarm setpoint in the positive direction.
DV-	Deviation Alarm -	Indicates a state in which the deviation between the process variable and the setpoint value exceeds the deviation alarm setpoint in the negative direction.
VEL+	Velocity Alarm +	Indicates a state in which the change amount of the input signal within a specified time exceeds the velocity limit alarm setpoint in the positive direction.
VEL-	Velocity Alarm -	Indicates a state in which the change amount of the input signal within a specified time exceeds the velocity limit alarm setpoint in the negative direction.
MHI	Output High Alarm	Indicate a state in which the output signal almost exceeded the output high-limit value. The actual output is limited to the output high-limit value.
MLO	Output Low Alarm	It indicates a state in which the output signal almost fell below the output low-limit value. The actual output is limited to the output low-limit value.
CNF	Connection Failure Alarm	Indicates a state in which a block mode of the function block in the I/O connection destination is in the out of service (O/S) mode. This alarm controls a temporary out of service state due to maintenance, and indicates a function block which is still in operation. Normally, IOP or OOP occurs simultaneously.

Process Alarm

Display sequence number	Tag mark	Date/Time alarm message occurred	Tag name	Annunciator message	Tag comment	Alarm status	Present process variable	Engineering unit symbol	Present alarm status
1	■	12/12 10:18:15	FIC005	Flowrate 5	NR	33.0M3/H	NR		
2	□	12/12 10:18:15	FIC005	Flowrate 5	LO	33.0M3/H	NR		
3	■	12/12 10:18:05	Tank level error	NR					
4	■	12/12 10:17:45	FIC005	Flowrate 5	LO				
5	■	12/12 10:17:45	Tank level error	ALM					
6	■	12/12 10:17:45	Tank temp error	ALM					
7	□	12/12 10:17:15	Pump fail		NR				
8	□	12/12 10:17:15	Pump fail		ALM				
9	■	12/12 10:17:05	FIC004	Flowrate 4	DV+	50.0M3/H	DV+		
10	■	12/12 10:16:45	FIC003	Flowrate 3	HH	12.0M3/H	HH		

System Alarm Window



The diagram illustrates the layout of the System Alarm Window. It features a table with seven rows of alarm data. Labels with arrows point to specific parts of the table: 'Display sequence number' points to the first column; 'System alarm mark' points to the diamond symbol in the second column; 'Time the alarm occurred' points to the date and time fields in the third and fourth columns; 'System alarm message' points to the text in the fifth column; and 'System alarm message number' points to the alphanumeric code in the second column.

Display sequence number	System alarm mark	System alarm message number	Time the alarm occurred	System alarm message
1	◆	0823	01/14 03:20:01	Printer EPSON MJ-930C Offline
2	◆	0502	01/14 02:31:02	DataBase Equalize Complete
3	◆	0510	01/14 02:30:57	Data Base Download from YOKOGAWA File = CIsDef.odt
4	◆	0510	01/14 02:30:57	Data Base Download from YOKOGAWA File = TrDspAssign.odt
5	◆	0510	01/14 02:30:57	Data Base Download from YOKOGAWA File = TrAssign.odt
6	◆	0502	01/14 02:29:33	DataBase Equalize Complete
7	◆	0510	01/14 02:29:23	Data Base Download from YOKOGAWA File = CIsMng.odt



Button	Description
	Specifies whether to display or hide the Toolbox.
	- When the pin is active The Toolbox is always visible. The Toolbox does not close without closing operation. - When the pin is not active The Toolbox closes after a single button operation.
	Calls up the System Status Overview window.
	Calls up the Help dialog for the active operation and monitoring window.
	Calls up the Process Alarm window.
	Calls up the Operation Guide window.
	Calls up the Graphic window with control attributes.
	Calls up the Tuning window.
	Calls up the Trend window.
	Calls up the Graphic window with graphic attributes.
	Calls up the Process Report window.
	Calls up the Historical Message Report window.

Toolbox



Button	Description
	Calls up the hierarchy window of the active window in ascending order and in the order specified in the window hierarchy definition.
	Calls up the upper windows of the active window.
	Calls up the hierarchy window of the active window in descending order and in the order specified in the window hierarchy definition.
	Calls up the Graphic window that has the process overview attribute.
	Saves the currently displayed operation and monitoring window group as a dynamic window set. An "M" follows the name of the operation and monitoring windows saved as a window set if it is a primary window and "S" follows if it is an auxiliary window.
	Deletes the currently set dynamic window set.
	Calls up the image window.
	Switches the active window in sequence, when multiple operation and monitoring windows are displayed.
	Calls up the active operation and monitoring window in a large size window.
	Calls up the active operation and monitoring window in a medium size window.